

DENIAL WORKFLOW MANAGEMENT

AR Follow-up Reporting Requirements

Functional and developer specification for operational monitoring, analyst productivity, workload, follow-up compliance, escalations, outcomes, and SLA reporting

Document control

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Implementation principle: Historical activity reporting and current operational backlog reporting use different grains. The application must not calculate current overdue workload from every historical note or status event.

1. Purpose and Scope

This specification defines the functional, data, calculation, presentation, security, export, and drill-down requirements for AR follow-up reporting within the Denial Workflow Management application. The report suite is intended to support daily operational control, analyst productivity review, workload balancing, management progress monitoring, escalation governance, closure analysis, and operational SLA tracking.

1.1 Business questions addressed

- What meaningful follow-up work was performed, and what changed?
- How much distinct work did each analyst complete without inflating results through repeated notes?
- Is current denial workload distributed appropriately by volume, balance, age, and complexity?
- Which appeals, rebills, write-offs, and other actions were completed, and when?
- Which follow-ups are due, overdue, completed on time, or rescheduled?
- How is denial inventory progressing through Open, Active, Pending, Escalated, and Closed buckets?
- How quickly are escalations answered and rework completed?
- How did closed items end, and which outcomes are operational versus financially verified?
- Are configured operational SLAs being met?

1.2 Report catalog

ID	Report	Primary management question
RPT-01	AR Follow-up Activity Detail	Provide an auditable record of every meaningful follow-up activity performed on a claim or denial line item.
RPT-02	AR Analyst Productivity Summary	Measure analyst output, workflow progression, and action completion without rewarding repeated notes or unnecessary status changes.
RPT-03	AR Analyst Workload and Capacity	Show whether denial workload is distributed appropriately and identify analysts carrying high-risk, stale, or complex backlogs.
RPT-04	Action Completion	Report when operational actions such as appeals, rebills, write-offs, documentation submissions, and payer follow-ups were completed.
RPT-05	Follow-up Due and Compliance	Provide the daily AR control plan and measure whether payer, escalation, and documentation follow-ups are completed on time.
RPT-06	Denial Work Progress by Classification and Action	Monitor the overall movement of denial inventory through stable reporting buckets by denial classification, action, task, and analyst.
RPT-07	Escalation Response and Rework	Measure clarification delays, manager response, escalation aging, and work returned to analysts for rework.
RPT-08	Closure and Outcome	Show how completed follow-up work ended, separating operational closure from verified financial or payer outcomes.
RPT-09	Operational SLA	Monitor whether key workflow milestones are completed within configurable operational service-level targets.

2. Common Reporting Architecture

2.1 Required logical data entities

Entity	Required content
Claim	Claim identifier, client/lab, payer, dates, status, balances, assignment context.
Denial Line Item	Line identifier, CPT, denial reason/classification, action/task, line status, line balance.
Assignment History	Assigned analyst/manager, effective date/time, assigned by, reassignment reason, prior assignee.
Activity Event	Immutable qualifying note, contact, status, schedule, escalation, completion, or rework event.
Status History	Previous/new operational status, reporting bucket, effective timestamp, user, source.
Action/Task Event	Action/task lifecycle, completed flag, completion timestamp, completion note, source.
Follow-up Schedule History	Original/revised due dates, category, revisions, reasons, completion linkage.
Escalation Cycle	Raised, due, response, rework, re-escalation, and resolution events.
Financial Snapshot	Charge and outstanding balance at defined snapshot times.
Outcome Evidence	Adjudication, payment, adjustment, appeal/rebill result, evidence source, verification state.
SLA Configuration and Instance	Versioned SLA rules and per-item measurement results.
Reporting Snapshot	Point-in-time inventory used for historical progress and workload reporting.

2.2 Event dates and business dates

Date	Definition	Primary use
Activity Date	Activity or note created date/time.	Activity reporting and claims worked.
Next Follow-up Date	Current scheduled due date.	Current work queue and due status.
Original Follow-up Date	First due date for the schedule instance.	Reschedule audit and historical compliance.
Action Completed Date	Timestamp when action is marked complete.	Appeal Date, Rebill Date, Write-off Completion Date, etc.
Escalated Date	Escalation cycle created timestamp.	Escalation aging and response SLA start.
Response Date	Manager response timestamp.	Response time and escalation status.
Rework Assigned/Completed	Rework cycle timestamps.	Rework turnaround and rate.
Closed Date	Valid closure event timestamp.	Closure volume and cycle time.
Outcome Verification Date	Date financial/payer outcome was verified.	Success and recovery metrics when available.

2.3 Standard reporting dimensions

- Client/Lab
- Date and reporting period

- AR Analyst, AR Manager, team
- Payer
- Denial classification and denial reason
- Action category and task
- Operational workflow status
- Stable reporting status bucket
- AR aging bucket
- Follow-up category and due status
- Priority
- Claim versus denial-line grain
- Update source
- Outcome verification state
- SLA type and result

2.4 Stable reporting status buckets

Reporting Bucket	Example Operational Statuses	Definition
Open	Open, Assigned, Not Started	Work exists but meaningful follow-up has not begun.
Active Follow-up	Follow-up Required, Appeal/Rebill in Process	Analyst is actively working the item.
Pending	Pending Payer Response, Pending Documentation	Progress depends on an external response or document.
Escalated	Escalated to AR Manager, Account Manager, or Client	Clarification/decision is required.
Closed	Completed, Resolved, Closed, Write-off Completed	Operational workflow cycle is closed.

Configuration requirement: Operational statuses must map to exactly one stable reporting bucket using effective-dated configuration. Report code must not embed status names.

2.5 Standard metric definitions

Term	Definition
Qualifying Activity	A work note, workflow status update, action/task completion, follow-up schedule update, escalation, response, or rework event. Passive navigation is excluded.
Claim Worked - Daily	Distinct claim + analyst + activity date with at least one qualifying activity.
Claim Worked - Period	Distinct claim + analyst with at least one qualifying activity within the selected period.
Outstanding Balance Worked	Balance counted once per distinct claim/analyst/activity date unless the report explicitly uses line-item grain.
Current Overdue	Latest open workflow item whose current next follow-up date is earlier than the report as-of date.
Historical Past-Due Schedule	A schedule instance whose due date is earlier than the as-of date; it is not automatically a current backlog item.

Term	Definition
Action Completed	A valid completion event with completed flag, completed user, and completion timestamp.
Operational Closure	A valid closure event. It does not imply payment or successful payer outcome.
Verified Financial Outcome	Outcome backed by accepted adjudication/payment evidence and verification status.

2.6 Common controls and presentation requirements

- **Filters** - All reports support authorized client/lab scope and report-specific filters. The applied-filter summary must remain visible and be included in exports.
- **As-of context** - Every report displays as-of date/time, generated-by user, generation timestamp, report run ID, and data refresh timestamp.
- **Grain labels** - Claim counts, denial-line counts, activity-event counts, action counts, and schedule counts must be clearly labeled and never presented as interchangeable totals.
- **Drill-down** - Every KPI and aggregate row opens the underlying filtered detail and retains the clicked context.
- **Export** - Excel export contains applied filters, summary totals, detail rows, run ID, generation metadata, and reconciliation totals.
- **Saved views** - Users can save filter, sort, grouping, and selected-column preferences within their authorization scope.
- **Role-based visibility** - Restricted internal notes and escalation content are excluded from unauthorized client-facing views and exports.
- **Empty state** - Reports state whether no records matched filters, data is unavailable, or a metric is not yet supported.

3. Report-Level Requirements

3.1 RPT-01 - AR Follow-up Activity Detail

Purpose and business question: Provide an auditable record of every meaningful follow-up activity performed on a claim or denial line item. It answers what work was performed, by whom, when, on which item, and what changed as a result.

Primary audience: AR Analyst (own authorized activities), AR Manager, Account Manager; restricted client-facing view without internal-only notes.

Report grain: One row per qualifying activity event per claim or denial line item.

Description: The report uses immutable note and workflow event history. It preserves every qualifying activity, while providing a Latest Activity Only option for operational review. Opening a page or viewing a claim does not create a qualifying activity.

Filters and dimensions

- Lab/client
- Activity date and time range
- AR analyst and AR manager

- Denial classification
- Action category and task
- Payer
- Workflow status and reporting status bucket
- AR aging bucket
- Follow-up due status
- Escalation status
- Update source (application or Excel upload)
- Latest Activity Only

Summary measures and formulas

Measure	Formula / Logic	Interpretation
Activity events	COUNT(activity_event_id)	All qualifying event rows after filters.
Distinct claims worked	COUNT DISTINCT (claim_id, analyst_id, activity_date)	A claim counts once per analyst per activity date.
Distinct denial lines worked	COUNT DISTINCT (line_item_id, analyst_id, activity_date)	Use when the user selects line-item grain.
Actions completed	COUNT(action_event_id WHERE completed = true)	Use completion timestamp as the action business date.
Outstanding balance worked	SUM one balance snapshot per distinct claim/analyst/activity date	Do not sum the same balance for every note event.
Escalations raised	COUNT(escalation_event_id WHERE event_type = Raised)	A response is a separate event.

Required detail columns

Column Group	Required Fields
Identification	Activity ID, Claim ID, Line Item ID, Encounter/Accession, CPT, DOS
Classification	Payer, denial classification, denial reason, reporting status bucket
Assignment	Assigned analyst, manager, assigned date/time, assigned by
Activity	Activity date/time, author, activity type, contact method, note text
Workflow change	Previous status, new status, status reason
Action	Action category, task, action completed Yes/No, action completed date/time
Follow-up	Next follow-up date, category, original due date, reschedule count
Escalation	Escalated Yes/No, reason, recipient, escalation ID
Financial	Original charge, outstanding insurance balance snapshot
Audit	Update source, upload batch/run ID, created/updated by

Business rules and caveats

- Count multiple notes as multiple activity events but one distinct claim-day worked when the claim, analyst, and activity date are the same.
- Latest Activity Only returns the most recent qualifying activity at the selected claim or line-item grain.
- Historical due dates in activity rows are audit snapshots; current overdue backlog must use the latest open workflow record.
- Internal notes must be excluded or masked for client-facing roles.

Drill-down path

Summary KPI -> Analyst / Classification / Action -> Claim -> Denial Line -> Complete activity timeline.

Acceptance criteria

- Three same-day notes on one claim by one analyst produce three activity events and one distinct claim-day worked.
- Every exported summary count reconciles to exported detail rows.
- Application and Excel-upload events are distinguishable in the audit columns.

3.2 RPT-02 - AR Analyst Productivity Summary

Purpose and business question: Measure analyst output, workflow progression, and action completion without rewarding repeated notes or unnecessary status changes.

Primary audience: AR Analyst (own summary), AR Manager, Account Manager. Client Manager receives aggregated client-level views only when authorized.

Report grain: One row per analyst for the selected reporting period, with daily and claim-level drill-down.

Description: The report separates activity volume, distinct work coverage, workflow progression, and completed actions. Note count is supporting context, not the primary productivity measure.

Filters and dimensions

- Lab/client
- Reporting period
- Analyst and manager
- Denial classification
- Action/task
- Payer
- AR aging bucket
- Priority
- Working-day calendar

Summary measures and formulas

Measure	Formula / Logic	Interpretation
Opening workload	Distinct open assigned claims at start of period	Point-in-time snapshot at period start.
New assigned	Distinct claims first assigned during period	Use assignment history.

Measure	Formula / Logic	Interpretation
Claims worked	COUNT DISTINCT (claim_id, analyst_id, activity_date) or period-level distinct claim	Display both daily productivity and period coverage.
Actions completed	COUNT completed actions by completed_at	Appeal, rebill, write-off, and configured tasks.
Closing workload	Distinct open assigned claims at end of period	Point-in-time snapshot at period end.
Average claims worked/day	Distinct claim-days worked / analyst working days	Exclude non-working days using the configured calendar.
Work coverage rate	Distinct workload claims worked / (opening workload + new assigned)	Cap only for display if reassignment creates denominator exceptions; retain raw values for audit.
Closure rate	Distinct claims closed / distinct claims worked	A closed claim must have a valid closure event.
No-activity rate	Open assigned claims with no qualifying activity / closing open workload	Define observation window in filter context.
Rework rate	Claims returned for rework / completed claims	Return event must be recorded separately.
Escalation rate	Distinct claims escalated / distinct claims worked	Avoid dividing escalation events by notes.

Required detail columns

Column Group	Required Fields
Analyst	Analyst ID/name, manager, team, active working days
Workload	Opening workload, newly assigned, reassigned in/out, closing workload
Activity	Distinct claims worked, denial lines worked, activity events, payer follow-ups, first follow-ups
Progression	Moved from Open, pending payer, pending documentation, escalated, rework assigned
Completion	Appeals, rebills, write-offs, other tasks, claims closed
Quality/control	Overdue current follow-ups, no next follow-up date, no-activity rate, rework rate
Financial	Balance worked, balance linked to completed actions, balance closed

Business rules and caveats

- A claim is worked when at least one qualifying activity is recorded.
- Count a claim once per analyst per day for daily productivity even when multiple notes exist.
- Reassignment must preserve attribution: activities remain with the performing analyst; workload snapshots follow the current assignee.
- Do not rank analysts solely on activity count because workload complexity and mix differ.

Drill-down path

Analyst -> Metric -> Activity date / status / action -> Claim -> Activity timeline.

Acceptance criteria

- Repeated notes do not inflate distinct claims worked.
- Opening + new assignments + reassigned in - reassigned out - closures reconciles to closing workload, subject to documented reopen adjustments.
- Working-day denominator matches the configured calendar.

3.3 RPT-03 - AR Analyst Workload and Capacity

Purpose and business question: Show whether denial workload is distributed appropriately and identify analysts carrying high-risk, stale, or complex backlogs.

Primary audience: AR Manager and Account Manager; analysts see their own workload.

Report grain: One row per analyst as of the selected date, with claim-level drill-down.

Description: This is a current-state inventory report, not an activity-history report. It uses the latest open assignment and workflow record for each claim or line item.

Filters and dimensions

- As-of date/time
- Lab/client
- Manager/team
- Analyst
- Denial classification
- Action/task
- Payer
- Status bucket
- AR aging
- Follow-up due status
- Priority

Summary measures and formulas

Measure	Formula / Logic	Interpretation
Open assigned claims	COUNT DISTINCT claim_id WHERE latest assignment is open	Current-state count.
Open denial lines	COUNT DISTINCT line_item_id WHERE latest line status is open	Line grain is separate from claim count.
Unworked assigned	Open claims with no qualifying activity since current assignment	Reset only when business rules define reassignment as a fresh work cycle.
Overdue follow-ups	Latest open items with next_follow_up_date < as_of_date	Never use every historical schedule row.
Average days since activity	AVG(as_of_date - latest_activity_date)	Exclude items with no activity and show them separately.
Workload units	SUM configurable work-type weight	Optional MVP enhancement; weights must be configurable.

Required detail columns

Column Group	Required Fields
Analyst	Analyst, manager, team, status/capacity flag
Inventory	Open claims, denial lines, outstanding balance
Work state	Unworked, active follow-up, pending payer, pending documentation, escalated
Control	Overdue, due today, no follow-up date, average days since last activity, oldest activity age
Mix	Aging, denial classification, action/task, payer, priority
Capacity	Configured capacity, workload units, utilization percentage, variance from team average

Business rules and caveats

- All counts use the latest open record as of the report timestamp.
- Raw counts and balance must be visible even if weighted workload is enabled.
- Default illustrative weights: payer follow-up 1.0, rebill 1.5, documentation follow-up 1.5, appeal 3.0, complex escalation/rework 3.0.
- Weights are configuration values, not fixed application constants.

Drill-down path

Team -> Analyst -> Status / aging / work type -> Claim -> Current assignment and activity timeline.

Acceptance criteria

- Closed items are excluded from current workload.
- Every analyst total reconciles to the open assignment detail.
- Changing a configured work weight recalculates workload units without changing raw counts.

3.4 RPT-04 - Action Completion

Purpose and business question: Report when operational actions such as appeals, rebills, write-offs, documentation submissions, and payer follow-ups were completed.

Primary audience: AR Analyst, AR Manager, Account Manager; authorized client roles may see action completion without internal notes.

Report grain: One row per completed action or task event.

Description: The date/time when the user marks the action complete becomes the business date for that action. The generic stored field can display as Appeal Date, Rebill Date, or Write-off Completion Date according to action type.

Filters and dimensions

- Completion date range
- Lab/client

- Analyst
- Action category
- Task
- Denial classification
- Payer
- Current status
- Update source

Summary measures and formulas

Measure	Formula / Logic	Interpretation
Completed actions	COUNT(action_event_id WHERE completed = true)	Group by action and task.
Completion balance	SUM balance snapshot associated with each completed action	Do not imply cash recovery.
Average days to completion	AVG(action_completed_at - assignment_or_task_created_at)	Start event must be defined by SLA configuration.
First-pass completion rate	Completed actions not returned for rework / completed actions	Requires rework linkage.

Required detail columns

Column Group	Required Fields
Identification	Claim ID, line item ID, encounter/accession, CPT
Assignment	Analyst, assigned date, task-created date
Classification	Payer, denial classification, action, task
Completion	Completed flag, completed by, completed date/time, action-specific displayed date
Workflow	Status at completion, current status, next follow-up date
Financial	Outstanding balance snapshot
Audit	Completion note, application/Excel upload, run/batch ID

Business rules and caveats

- An action completion event is immutable; later correction creates an audited amendment rather than silently replacing the timestamp.
- Appeal completed -> Appeal Date; Rebill completed -> Rebill Date; Write-off completed -> Write-off Completion Date; Medical records submitted -> Documentation Submission Date.
- Operational completion does not equal payer success or payment recovery.

Drill-down path

Action -> Task -> Analyst -> Completion date -> Claim -> Action event and supporting activity history.

Acceptance criteria

- Action-specific date labels display correctly.
- The action completion count reconciles to rows with a completion timestamp.
- Excel-upload completion events are validated and audited identically to application updates.

3.5 RPT-05 - Follow-up Due and Compliance

Purpose and business question: Provide the daily AR control plan and measure whether payer, escalation, and documentation follow-ups are completed on time.

Primary audience: AR Analyst (own due list), AR Manager, Account Manager.

Report grain: One row per follow-up schedule instance, plus a current open-item view using the latest schedule.

Description: The report supports both current work queues and historical compliance. Original and revised due dates must be retained so a user cannot move an overdue item into the future without audit visibility.

Filters and dimensions

- As-of date
- Due-date range
- Follow-up category
- Analyst/manager
- Lab/client
- Denial classification
- Action/task
- Payer
- Due status
- Priority
- Rescheduled Yes/No

Summary measures and formulas

Measure	Formula / Logic	Interpretation
Follow-ups due	COUNT schedule instances with due date in period	Use original schedule instance history.
Completed on time	COUNT completed_at <= effective_due_date	Define effective due date and reschedule policy.
Completed late	COUNT completed_at > effective_due_date	Show average late days.
Current overdue	Latest open items where next_follow_up_date < as_of_date	Current-state metric.
Compliance percentage	Completed on or before due date / total follow-ups due * 100	State treatment of cancelled schedules.
No follow-up scheduled	Open items whose status requires follow-up and current due date is blank	Status requirement comes from configuration.

Required detail columns

Column Group	Required Fields
Identification	Follow-up schedule ID, claim/line IDs, payer, denial classification
Ownership	Analyst, manager, assigned date
Schedule	Category, original due date, current/revised due date, revised by/date/reason, reschedule count
Completion	Qualifying activity ID, completed date/time, completed by, on-time/late status
Current control	Current workflow status, days overdue/due, priority, recommended next action
Financial	Outstanding balance snapshot

Business rules and caveats

- Due status: Overdue if current due date is before as-of and item remains open; Due Today if equal; Due Soon if within configured future window; Future otherwise.
- When a due date changes, preserve original date, revised date, user, timestamp, and reason.
- A qualifying completion activity must match the follow-up category; an unrelated note does not satisfy the schedule.
- Separate payer, escalation, and documentation follow-up compliance.

Drill-down path

Due status -> Category -> Analyst -> Claim -> Schedule history and qualifying completion event.

Acceptance criteria

- Rescheduling preserves the original due date and does not erase late-history evidence.
- Current overdue counts use latest open records only.
- Compliance numerator and denominator reconcile to schedule-instance detail.

3.6 RPT-06 - Denial Work Progress by Classification and Action

Purpose and business question: Monitor the overall movement of denial inventory through stable reporting buckets by denial classification, action, task, and analyst.

Primary audience: AR Manager, Account Manager, Client Manager; analysts may see their assigned scope.

Report grain: Aggregated by selected dimension and reporting bucket, with both claim/line counts and outstanding balance.

Description: This management report separates operational workflow status names from stable reporting buckets so configuration changes do not break trend reporting.

Filters and dimensions

- As-of date or snapshot period
- Lab/client
- Denial classification

- Action/task
- Analyst/manager
- Payer
- AR aging
- Reporting bucket
- Priority

Summary measures and formulas

Measure	Formula / Logic	Interpretation
Total inventory	Open-scope claim or line count and balance	Display grain selector or separate measures.
Open	Reporting bucket = Open	Examples: Open, Assigned, Not Started.
Active Follow-up	Reporting bucket = Active Follow-up	Examples: Follow-up Required, Appeal/Rebill in Process.
Pending	Reporting bucket = Pending	Examples: Pending Payer Response, Pending Documentation.
Escalated	Reporting bucket = Escalated	Manager, Account Manager, or Client escalation.
Closed	Reporting bucket = Closed	Completed, Resolved, Closed, Write-off Completed.
Overdue	Latest open item currently overdue	Overlay control, not a mutually exclusive bucket.

Required detail columns

Column Group	Required Fields
Group dimension	Denial classification / action / task / analyst / payer
Counts	Total, Open, Active, Pending, Escalated, Closed, Overdue
Balances	Balance for each corresponding count measure
Movement	Opening inventory, new assigned, moved forward, closed, reopened, closing inventory
Mix	Percentage of total count and balance

Business rules and caveats

- Status-to-reporting-bucket mapping is configuration data with effective dates.
- Claim and line-item counts must never be combined under one unlabeled total.
- Overdue is a cross-cutting flag and may coexist with Active, Pending, or Escalated.
- Snapshot reporting is required for historical trend; current tables alone cannot reconstruct past inventory accurately.

Drill-down path

Denial Classification -> Action Category -> Task -> Analyst -> Claim -> Denial Line.

Acceptance criteria

- Every operational status maps to exactly one reporting bucket for an effective date.
- Count and balance totals reconcile across buckets at the selected grain.
- Drill-down retains the clicked dimension and filter context.

3.7 RPT-07 - Escalation Response and Rework

Purpose and business question: Measure clarification delays, manager response, escalation aging, and work returned to analysts for rework.

Primary audience: AR Analyst (own escalations), AR Manager, Account Manager; selected client-facing escalation visibility by role.

Report grain: One row per escalation event/cycle, linked to response and rework events.

Description: Manager response and final claim resolution are separate milestones. An escalation can be responded to, returned for rework, re-escalated, and later resolved.

Filters and dimensions

- Escalated date range
- Response date range
- Escalation status
- Escalation reason
- Raised by analyst
- Escalated to
- Lab/client
- Denial classification
- Action/task
- Overdue status

Summary measures and formulas

Measure	Formula / Logic	Interpretation
Escalations raised	COUNT escalation cycles created in period	One claim may have multiple cycles.
Open escalations	Latest escalation status is open	Current-state measure.
Average response time	AVG(manager_response_at - escalated_at)	Use configured working hours/days if required.
Overdue response	Open escalation where response due date < as_of_date	Due date comes from SLA or explicit next follow-up.
Rework rate	Escalation cycles returned for rework / responded escalation cycles	Response alone is not resolution.
Repeat escalation rate	Claims with more than one escalation cycle / escalated claims	Define reporting-period boundary.

Required detail columns

Column Group	Required Fields
Identification	Escalation ID/cycle, claim/line IDs, encounter, CPT
Raised	Raised by, escalated date/time, reason, note, recipient
Follow-up	Response due/next follow-up, days awaiting response, overdue flag
Response	Responded by/date, response type, response note, recommended action
Rework	Rework required, returned date, assigned analyst, completed date, cycle time
Resolution	Escalation status, escalation closed date, final resolution
Financial	Outstanding balance snapshot

Business rules and caveats

- AR Manager response returns to the same assigned analyst unless a reassignment event is recorded.
- A response event does not automatically close the claim or escalation.
- Repeat escalation is counted by distinct escalation cycle IDs on the same claim.
- Restricted internal escalation notes must not appear in client-facing exports.

Drill-down path

Escalation status/reason -> Recipient / analyst -> Claim -> Full escalation-response-rework timeline.

Acceptance criteria

- Response time uses the correct raised and response timestamps.
- Returned-for-rework cycles preserve both original analyst and current assignee.
- Open escalation totals reconcile to latest escalation statuses.

3.8 RPT-08 - Closure and Outcome

Purpose and business question: Show how completed follow-up work ended, separating operational closure from verified financial or payer outcomes.

Primary audience: AR Manager, Account Manager, Client Manager; analysts see their closures.

Report grain: One row per closed claim or denial line item, according to the selected report grain.

Description: The report can launch with operational closure measures. Appeal success, rebill success, payment recovery, and dollars recovered remain conditional until reliable adjudication/payment data is integrated.

Filters and dimensions

- Closed date range
- Lab/client
- Closed by analyst

- Final action
- Final resolution
- Denial classification
- Payer
- AR aging
- Reopened Yes/No
- Outcome verification status

Summary measures and formulas

Measure	Formula / Logic	Interpretation
Claims/lines closed	COUNT DISTINCT closed item at selected grain	Closure event required.
Average days to closure	AVG(closed_at - assignment_at)	Define assignment cycle for reassigned/reopened items.
First-pass completion	Closed without rework or reopen / closed items	Requires event linkage.
Reopened rate	Closed items later reopened / closed items	Show reopened count separately.
Appeal success rate	Successful adjudicated appeals / adjudicated appeals	Deferred until verified outcome data.
Rebill success rate	Successful adjudicated rebills / adjudicated rebills	Deferred until verified outcome data.
Dollars recovered	SUM verified post-action payments attributable to closed item	Requires attribution and payment integration.

Required detail columns

Column Group	Required Fields
Identification	Claim/line IDs, encounter/accession, CPT, DOS
Classification	Payer, denial classification, final action/task
Closure	Closed date/time, closed by, final resolution, operational closure reason
Cycle	Assigned date, days to closure, activity count, rework cycles, reopened flag/date
Financial snapshot	Original outstanding balance, balance at closure, adjustment/write-off amount
Verified outcome	Payment received, appeal/rebill outcome, verification source/date, remaining balance

Business rules and caveats

- Label unverified results as Operational Completion, not Financial Success.
- A closed status must have a closure timestamp, user, and reason.
- Reopened items retain the original closure event and create a new workflow cycle.
- Financial outcome fields remain null/Not Available until reliable source integration exists.

Drill-down path

Final action/resolution -> Analyst / payer / classification -> Closed item -> Full lifecycle and outcome evidence.

Acceptance criteria

- Operational closure KPIs do not use unverified payment data.
- Reopened items are traceable to prior closure cycles.
- Verified success metrics only include records with accepted outcome evidence.

3.9 RPT-09 - Operational SLA

Purpose and business question: Monitor whether key workflow milestones are completed within configurable operational service-level targets.

Primary audience: AR Manager, Account Manager, Client Manager for authorized client-level SLA views.

Report grain: One row per SLA measurement instance per claim, line item, task, escalation, or rework cycle.

Description: SLA definitions are configuration records that identify the event that starts the clock, the event that stops it, the calendar, target duration, warning threshold, and applicable dimensions.

Filters and dimensions

- SLA measurement period
- SLA type
- Lab/client
- Analyst/manager
- Action/task
- Denial classification
- Payer
- Priority
- Met/Breached/In Progress
- Due window

Summary measures and formulas

Measure	Formula / Logic	Interpretation
SLA instances	COUNT measurement instances in scope	Separate completed and in-progress.
SLA met percentage	Completed within target / completed SLA instances * 100	In-progress items excluded from completed denominator but shown separately.
Breached	Completed late plus open items past due	Display completed breach and active breach separately.
Average turnaround	AVG(end_at - start_at)	Apply configured calendar.
Near breach	Open instance within warning threshold of due time	Threshold is configuration.

Required detail columns

Column Group	Required Fields
SLA identity	SLA instance ID, SLA type, configuration version
Item	Claim/line/task/escalation IDs, analyst, client
Clock	Start event/date, target duration, due date/time, warning threshold, calendar
Completion	End event/date, elapsed duration, pause duration, net duration
Result	Met/Breached/In Progress/Near Breach, breach amount
Context	Denial classification, action/task, payer, priority, status

Business rules and caveats

- Supported initial SLAs: first follow-up, appeal completion, rebill completion, payer follow-up, escalation response, and rework completion.
- Targets can vary by client/lab, action, task, denial classification, payer, and priority.
- Configuration uses effective dates and versioning so historical results are reproducible.
- Any pause-clock behavior must identify allowed pause statuses and preserve pause history.

Drill-down path

SLA type/result -> Client / analyst / action -> SLA instance -> Start/end/pause event evidence.

Acceptance criteria

- Each SLA result traces to explicit start and end events.
- Changing a target applies prospectively based on effective dates and does not rewrite historical SLA results.
- Active breaches and completed breaches are reported separately.

4. Cross-Report Functional Requirements

ID	Requirement	Specification
FR-001	Report run metadata	Every report displays report name, run ID, generated-by user, generation timestamp, as-of timestamp, data refresh timestamp, and applied filters.
FR-002	Reconciliation	Every summary metric must reconcile to the corresponding drill-down detail at the same grain and filter context.
FR-003	Status configuration	Status-to-reporting-bucket mappings are configurable, effective-dated, and auditable.
FR-004	Date configuration	All timestamps are stored consistently and displayed in the authorized client/lab timezone. Date-only and date-time fields are clearly distinguished.
FR-005	Excel export	Excel exports preserve filters, column order, typed dates/numbers, totals, and audit metadata.
FR-006	Role security	Analysts see authorized own scope; managers see authorized teams/clients; client-facing roles receive only authorized data and note categories.

ID	Requirement	Specification
FR-007	History preservation	Assignment, status, action, follow-up date revisions, escalation, closure, reopen, and outcome histories remain immutable and auditable.
FR-008	Current vs history	Current operational metrics use latest-state records; activity and compliance metrics use event/schedule history according to their definitions.
FR-009	Claim/line grain	The UI and export clearly identify whether metrics represent claims, denial lines, activities, actions, schedules, or escalation cycles.
FR-010	Drill-through	Clicking a KPI, chart segment, or aggregate row opens the filtered underlying data and provides a path to the claim timeline.
FR-011	Unavailable metrics	Metrics that depend on unavailable adjudication/payment data display Not Available and state the missing dependency.
FR-012	Audit source	Updates made through Excel upload identify source file/batch/run and pass the same validation and authorization rules as UI updates.

5. Non-Functional and Data Quality Requirements

ID	Area	Requirement
NFR-001	Performance	Summary pages should return within 5 seconds for normal authorized scope; large detail exports may run asynchronously with visible status.
NFR-002	Security and PHI	Apply least privilege, secure transport, audit logging, and role-based masking/exclusion of restricted notes and identifiers.
NFR-003	Auditability	Each metric must be traceable to source records, rule version, report run ID, and as-of context.
NFR-004	Data freshness	Display the last successful workflow/data refresh time and flag stale or partial loads.
NFR-005	Data quality	Validate missing identifiers, orphaned events, unmapped statuses, invalid date sequences, duplicate completion events, and negative/invalid balance snapshots.
NFR-006	Accessibility	Keyboard-accessible filters, tables, drill-down controls, readable status labels, and non-color-only indicators.
NFR-007	Responsive design	Desktop-first management workspace with functional tablet behavior; wide tables support sticky identifiers and horizontal scrolling.
NFR-008	Versioning	Metric definitions, bucket mappings, SLA rules, and workload weights are versioned and effective-dated.

6. Implementation Priority

Phase	Reports	Rationale / Dependency
MVP 1	RPT-01 Activity Detail; RPT-02 Productivity; RPT-03 Workload; RPT-05 Follow-up Due; RPT-06 Denial Progress	Uses existing note history, assignment, status, action completion, current follow-up date, classification, and balance data.

Phase	Reports	Rationale / Dependency
MVP 2	RPT-04 Action Completion; RPT-07 Escalation/Rework; RPT-09 Operational SLA	Requires consistent event timestamps, escalation/rework linkage, and versioned SLA configuration.
Enhancement	RPT-08 Closure and Outcome financial-success measures	Requires reliable adjudication/payment integration and outcome attribution. Operational closure can launch earlier.

7. Open Configuration Decisions

- Confirm whether productivity is displayed primarily by distinct claim-day, distinct period claim, or both. Recommendation: show both, with claim-day for daily productivity and period claim for coverage.
- Define working-day calendars and holidays by client/team for productivity and SLA calculations.
- Confirm statuses that require a next follow-up date and the default Due Soon window.
- Approve workload-complexity weights and whether weighted capacity is part of MVP or a later enhancement.
- Define allowed pause statuses and pause-clock behavior for SLA measurements.
- Confirm client-facing note categories and fields that must be masked or excluded.
- Define accepted financial outcome evidence and attribution rules before enabling appeal/rebill success and dollars recovered.

8. Revision History

Version	Date	Status	Summary
1.0	2026-09-02	Draft for Developer Review	Initial consolidated requirements for nine AR follow-up report templates, formulas, data points, dimensions, columns, drill-downs, caveats, and acceptance criteria.